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(12) **United States Patent**
Itabashi

(10) **Patent No.:** **US 12,411,439 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **IMAGE FORMING APPARATUS INCLUDING
PROCESS CARTRIDGE, AND TONER
CARTRIDGE HAVING TONER SHUTTER
PIVOTALLY MOVABLE IN A STATE WHERE
TONER CARTRIDGE IS ATTACHED TO
PROCESS CARTRIDGE**

(58) **Field of Classification Search**
CPC G03G 15/0863; G03G 15/0865; G03G
15/0886; G03G 21/1652; G03G 21/1814;
G03G 21/1867

(Continued)

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Dec. 3, 2021 (JP) 2021-196723

(51) **Int. Cl.**
G03G 15/00 (2006.01)
G03G 15/08 (2006.01)

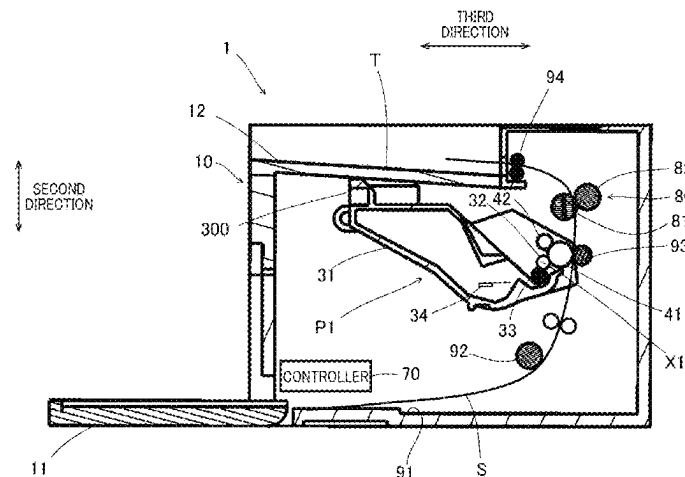
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(52) **U.S. Cl.**
CPC **G03G 15/0886** (2013.01); **G03G 15/0863**
(2013.01); **G03G 15/0865** (2013.01);
(Continued)

(57) **ABSTRACT**

An image forming apparatus includes: a main housing including a main electrical contact; a process cartridge attachable to and detachable from the main housing; and a toner cartridge attachable to and detachable from the process cartridge attached to the main housing. The process cartridge includes: a first process electrical contact configured to contact the main electrical contact; and a second process electrical contact. The toner cartridge includes: a toner housing having a toner outlet port and configured to accommodate toner therein; a toner memory including a toner electrical contact; and a toner shutter. In a state where the toner cartridge is attached to the process cartridge, the toner electrical contact is in contact with the second process

(Continued)



electrical contact; and the toner shutter is pivotally movable relative to the second process electrical contact and the toner electrical contact to open and close the toner outlet port.

15 Claims, 11 Drawing Sheets

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G03G 21/16 (2006.01)
G03G 21/18 (2006.01)
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 CPC **G03G 21/1652** (2013.01); **G03G 21/1814**
 (2013.01); **G03G 21/1867** (2013.01)
- (58) **Field of Classification Search**
 USPC 399/90, 107, 110, 111
 See application file for complete search history.

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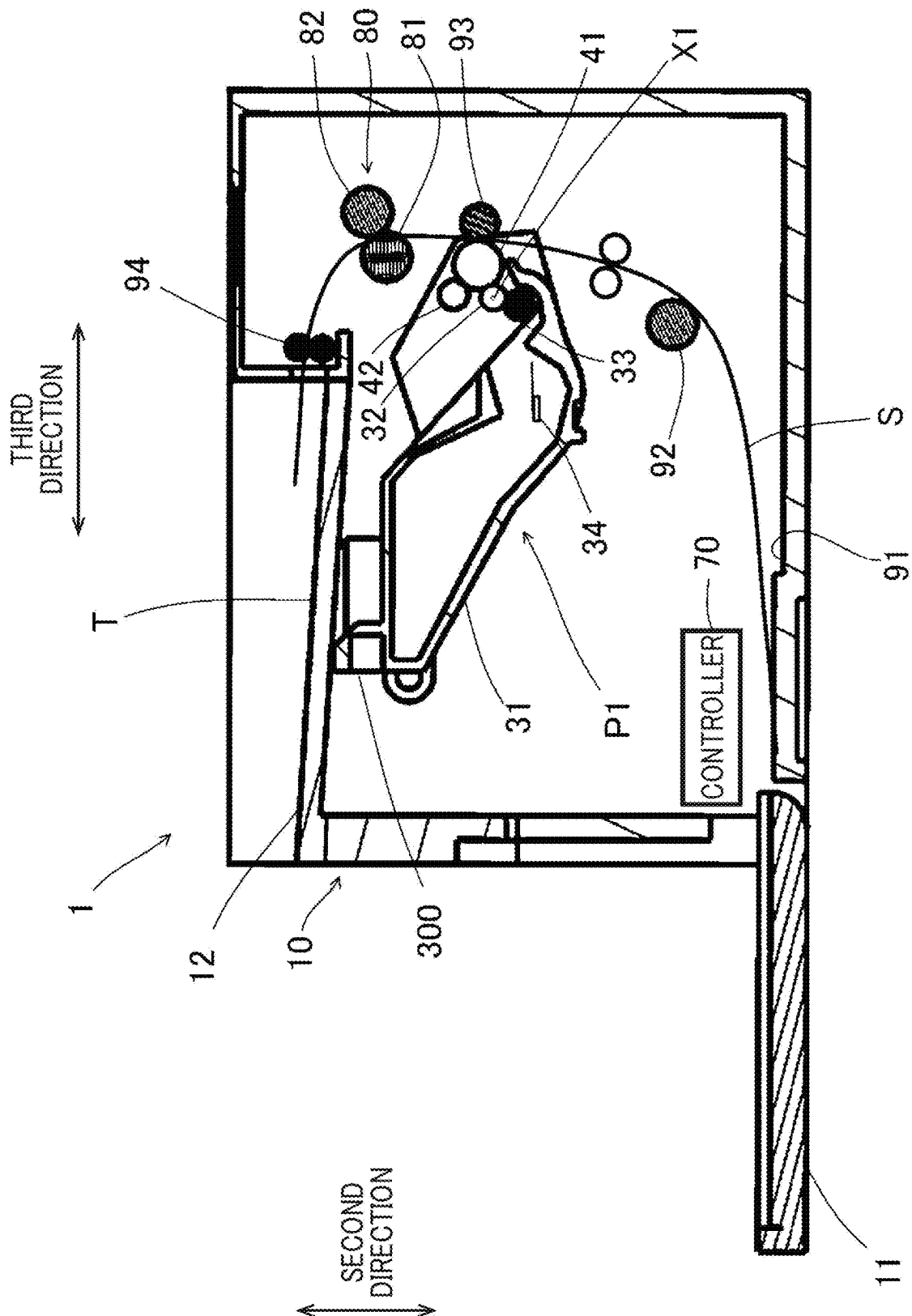


FIG. 2A

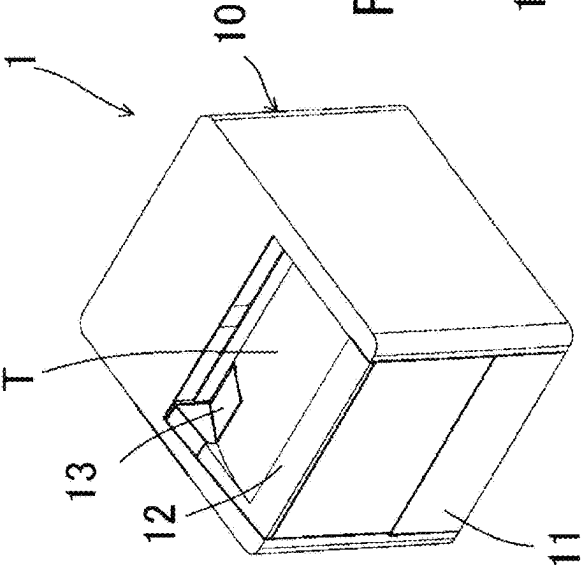


FIG. 2B

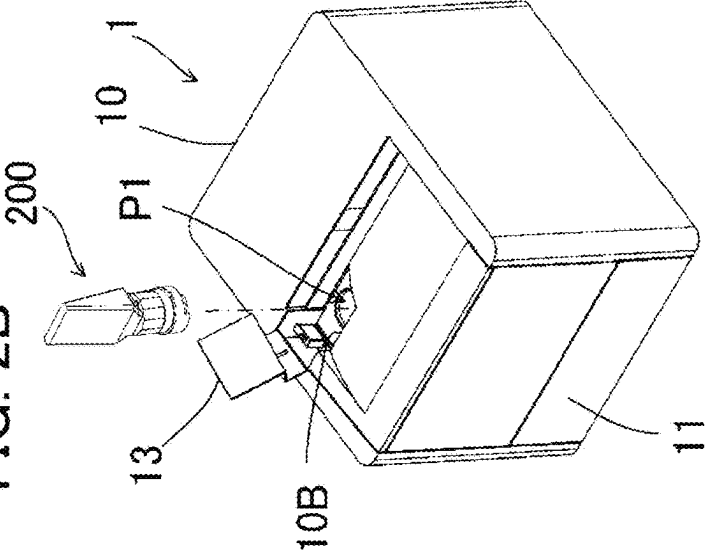
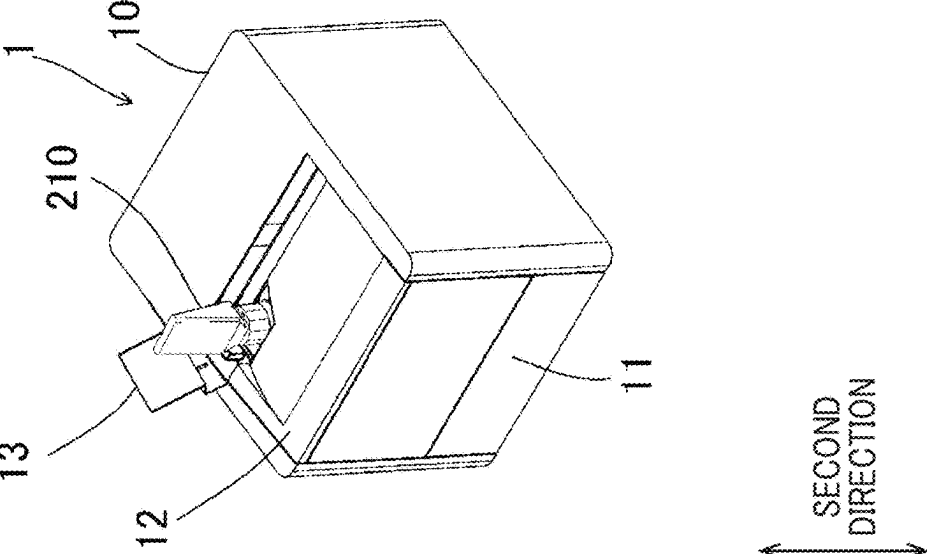
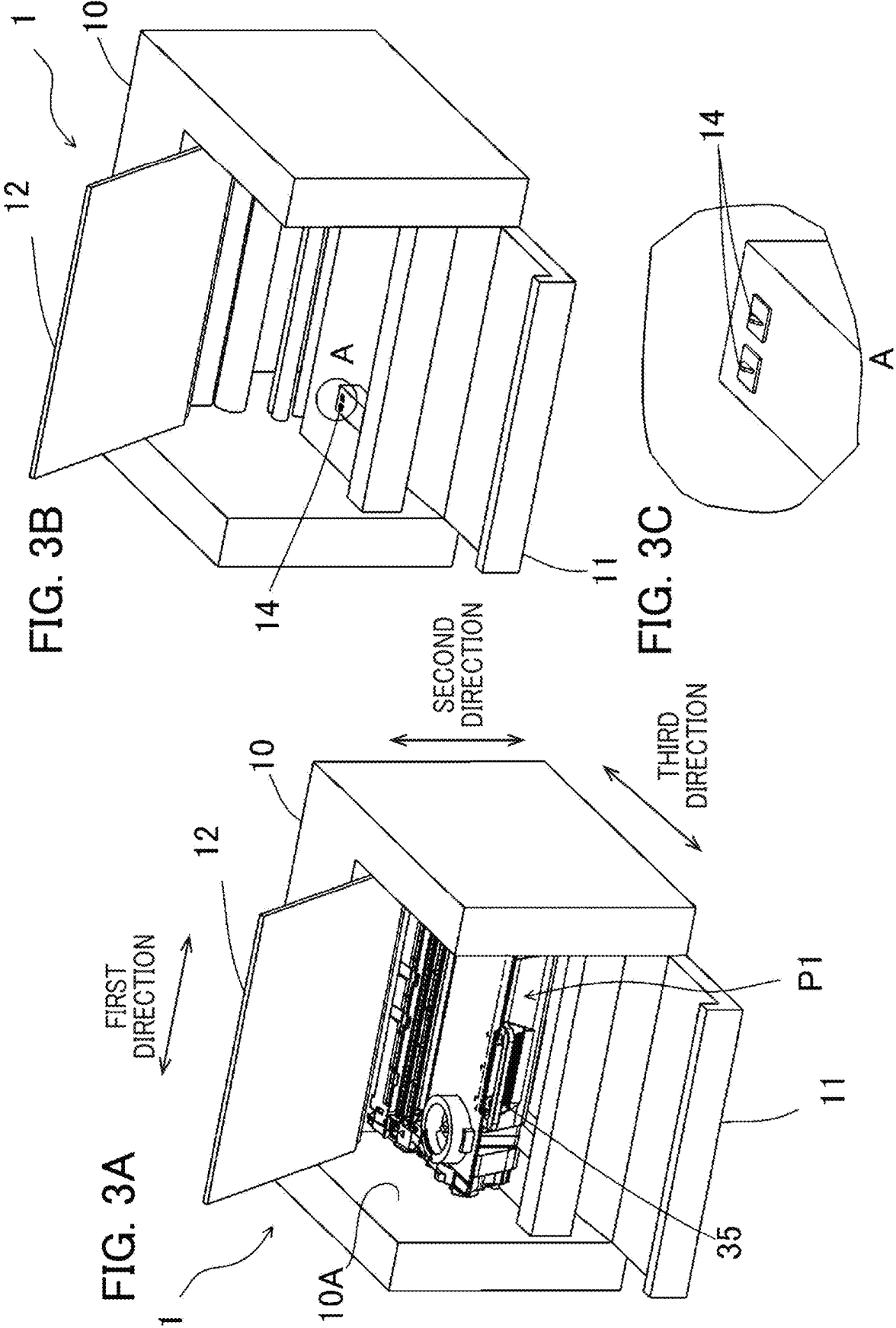


FIG. 2C





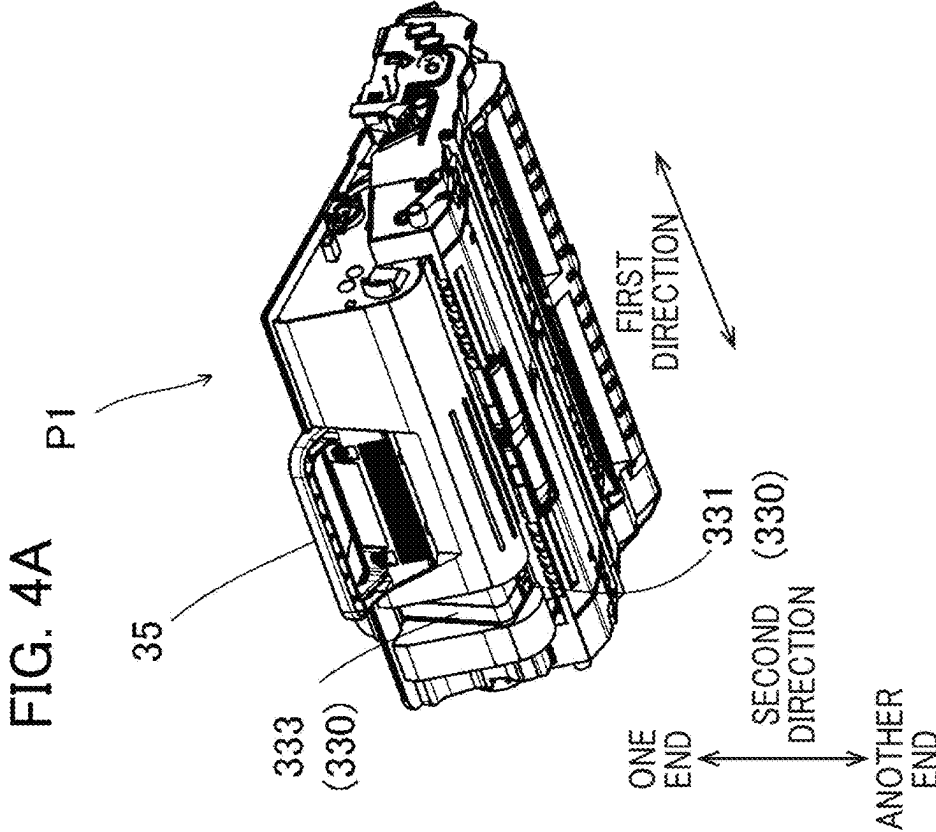
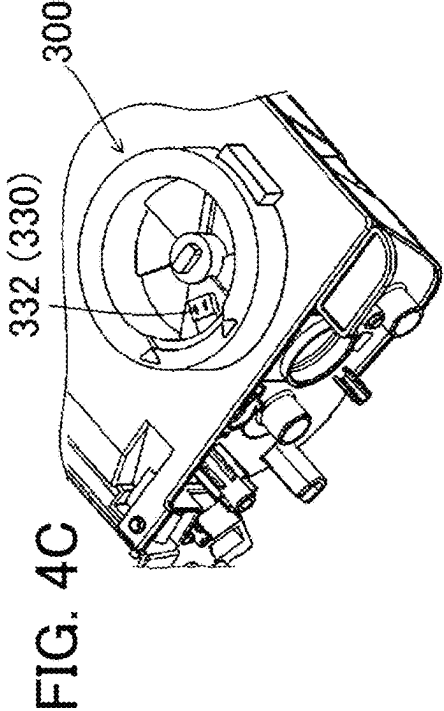
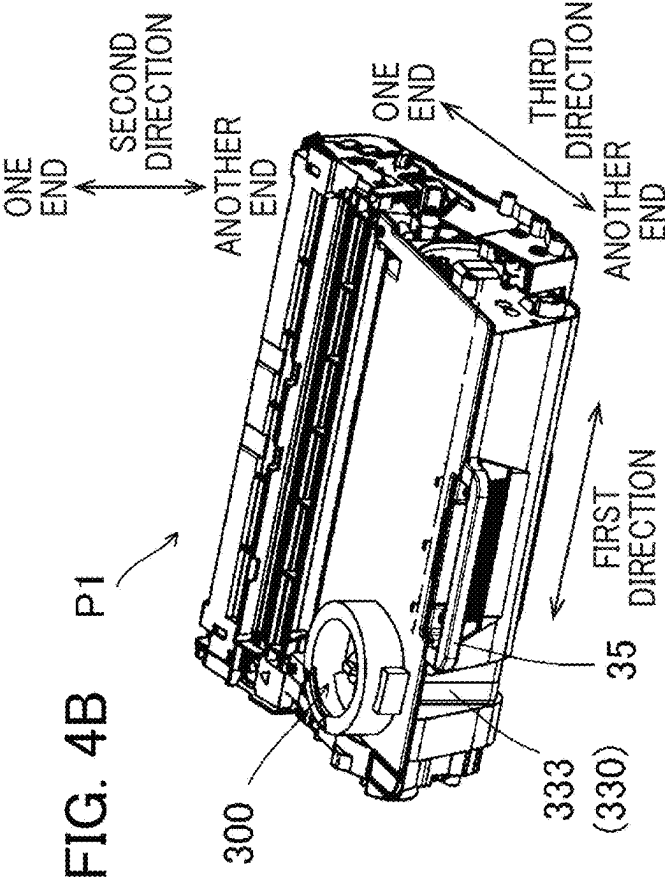


FIG. 5A

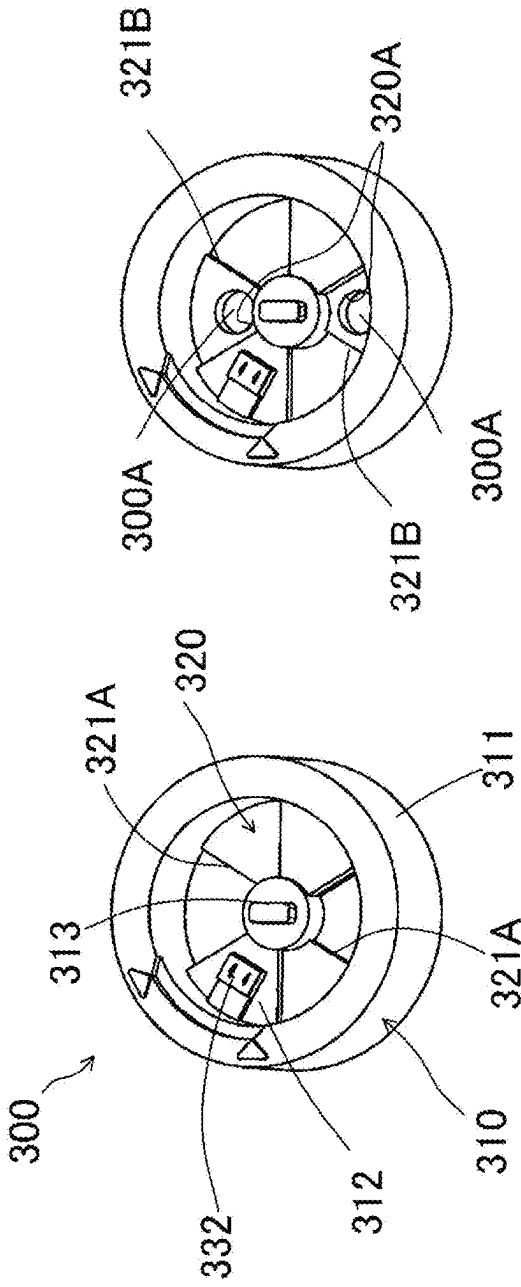


FIG. 5B

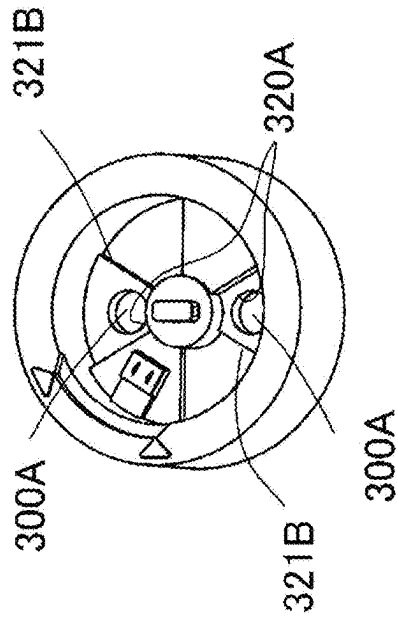


FIG. 5C

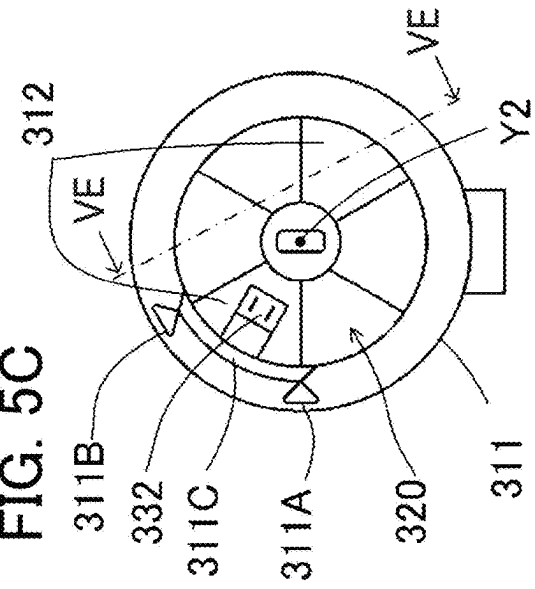


FIG. 5D

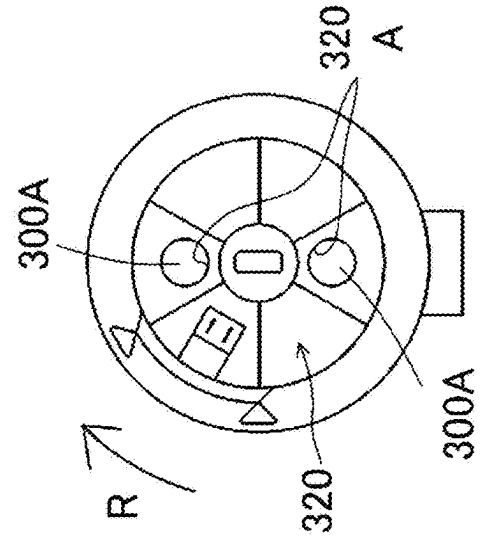


FIG. 5E

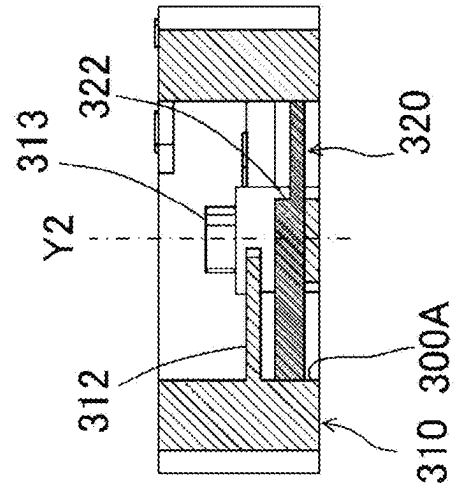


FIG. 6B

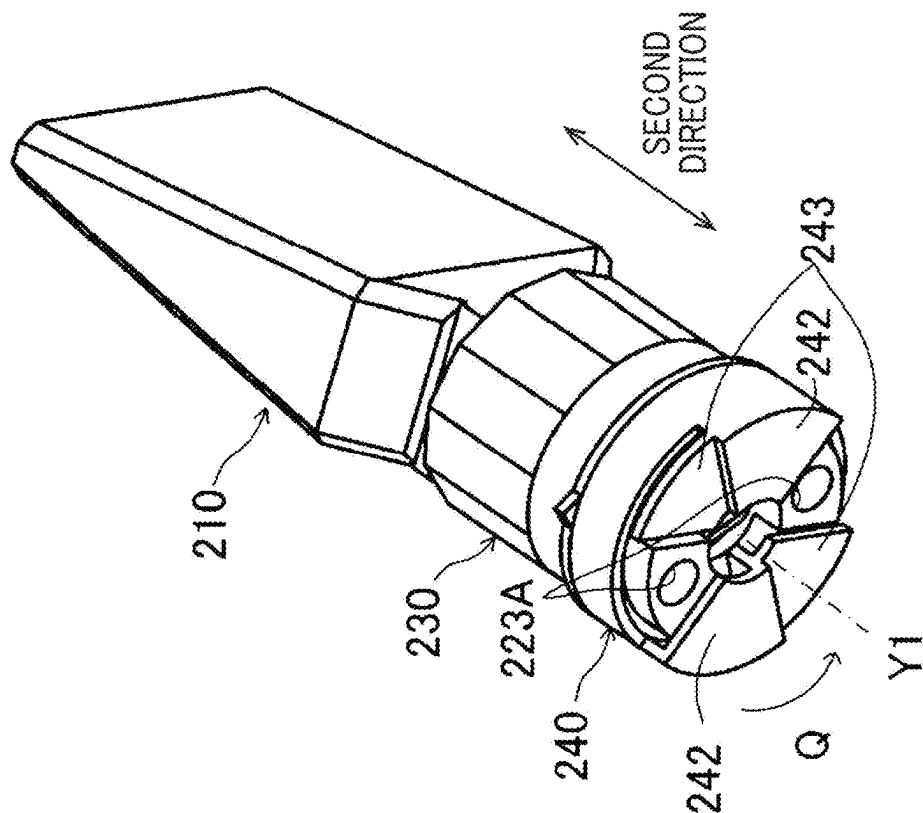


FIG. 6A

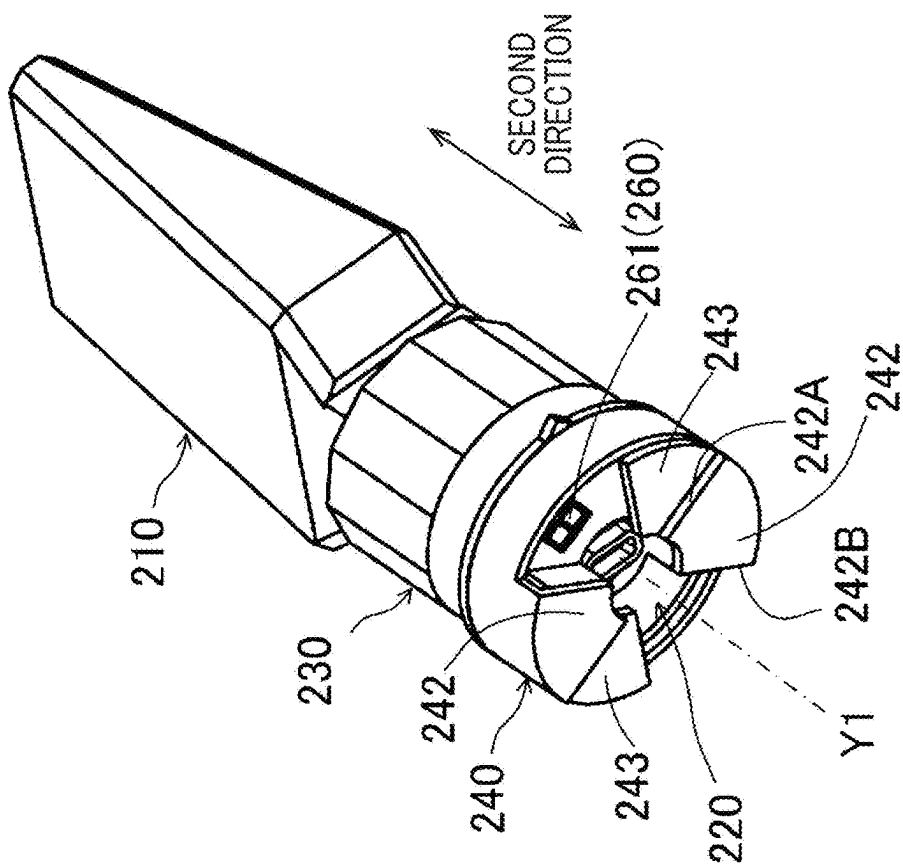


FIG. 7A

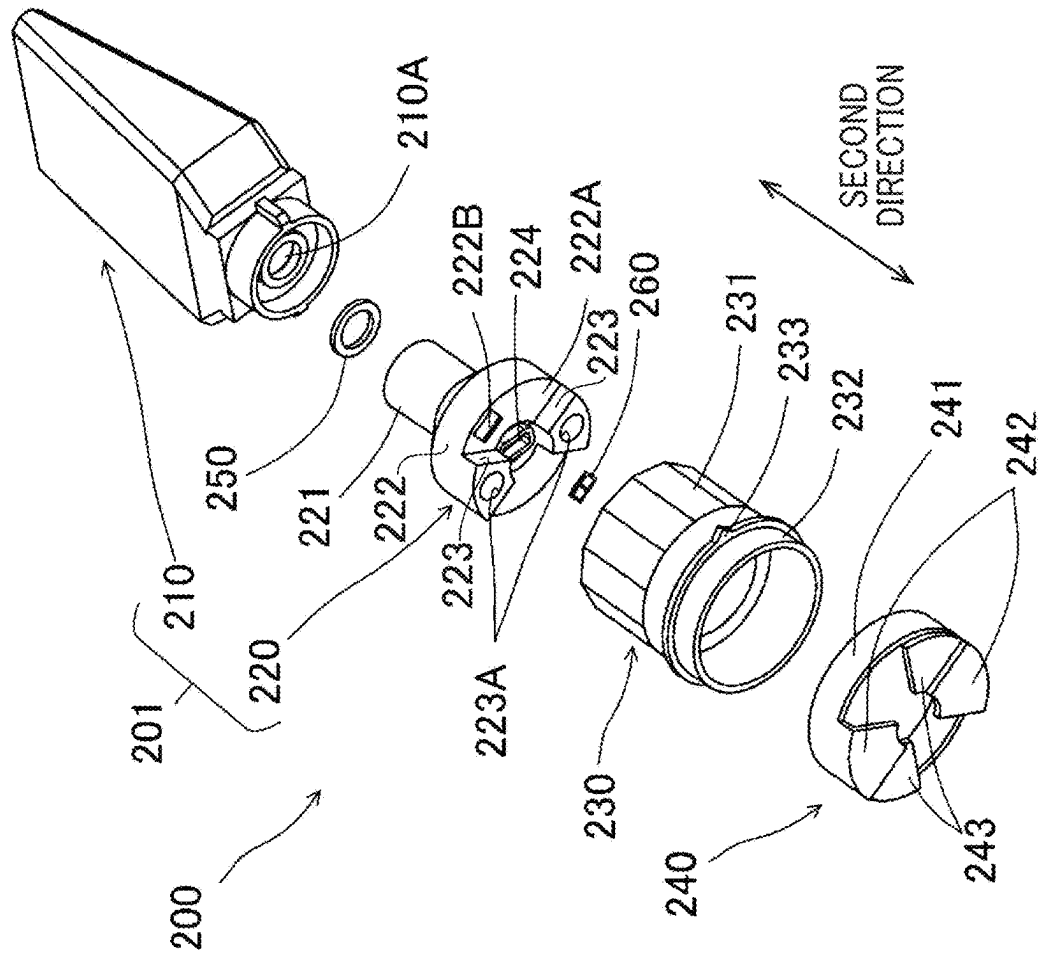


FIG. 7B

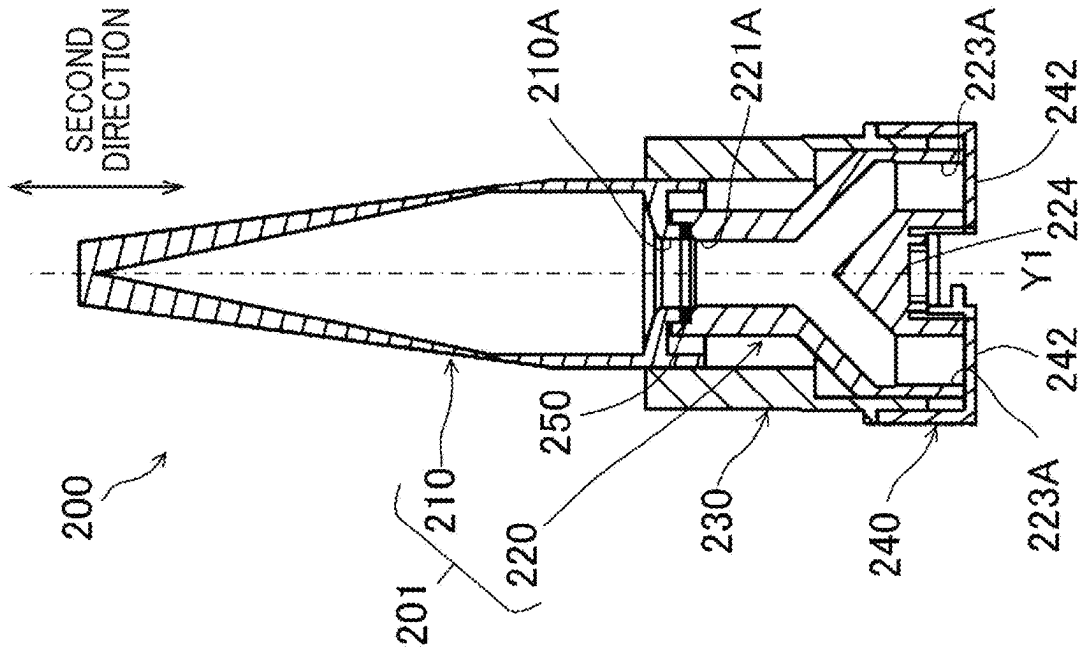


FIG. 8A

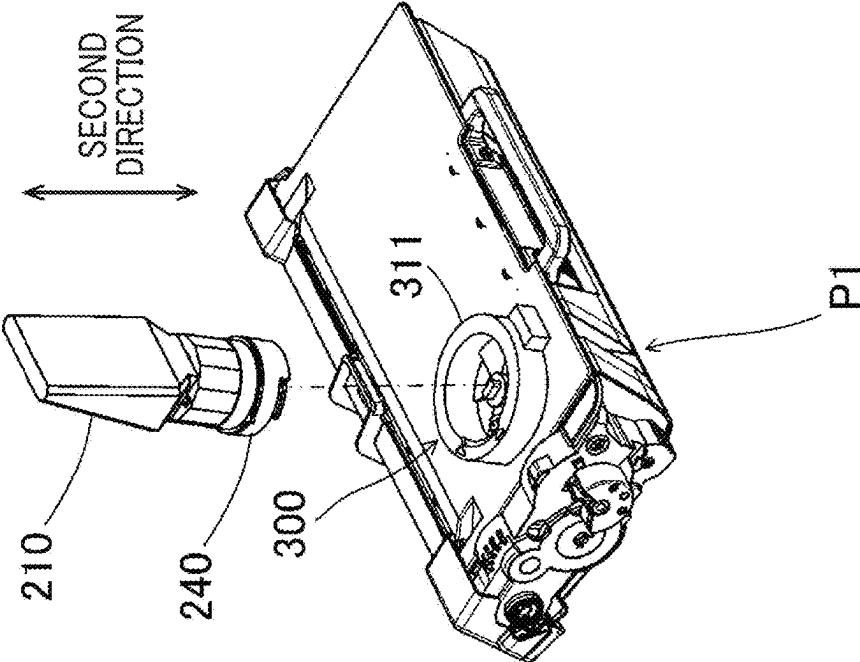


FIG. 8B

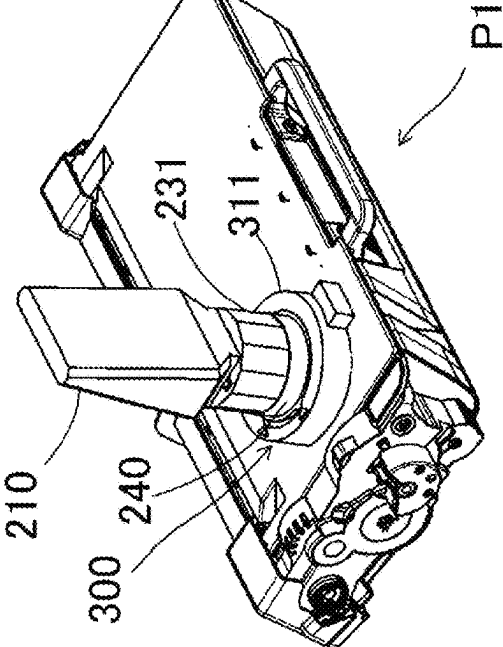


FIG. 8C

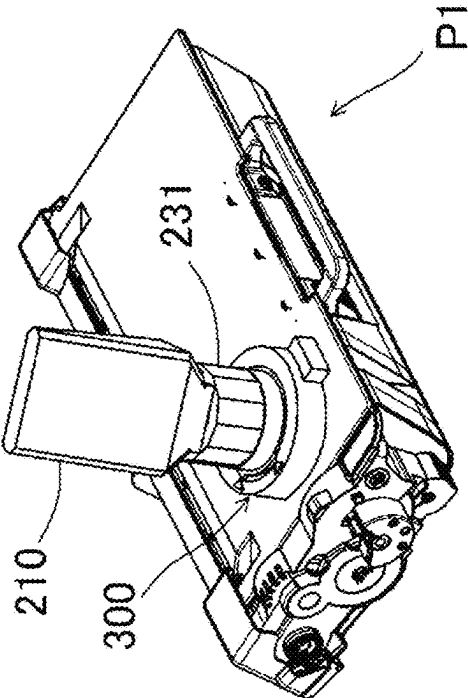


FIG. 9B

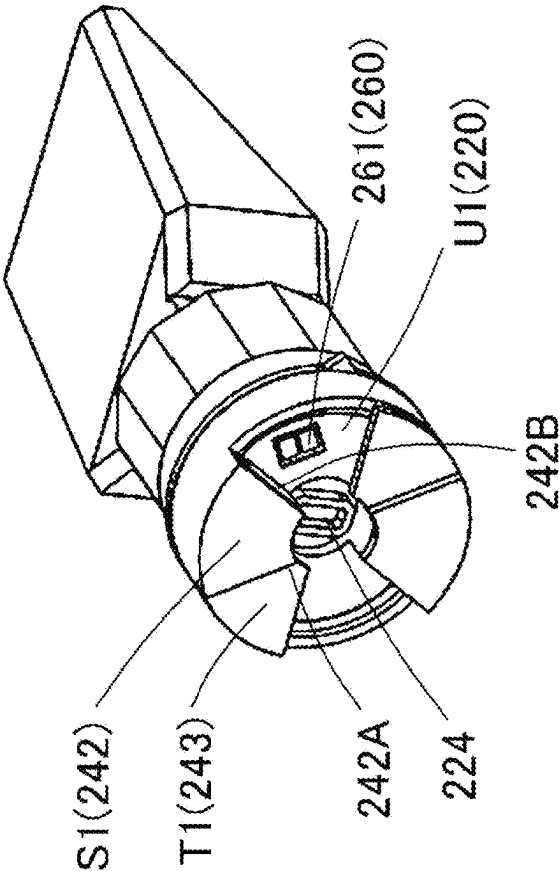


FIG. 9A

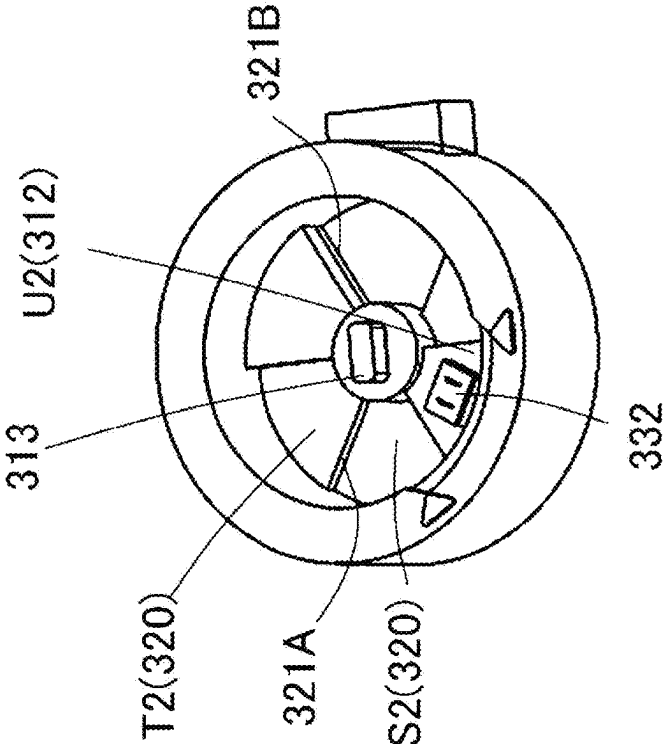


FIG. 10A

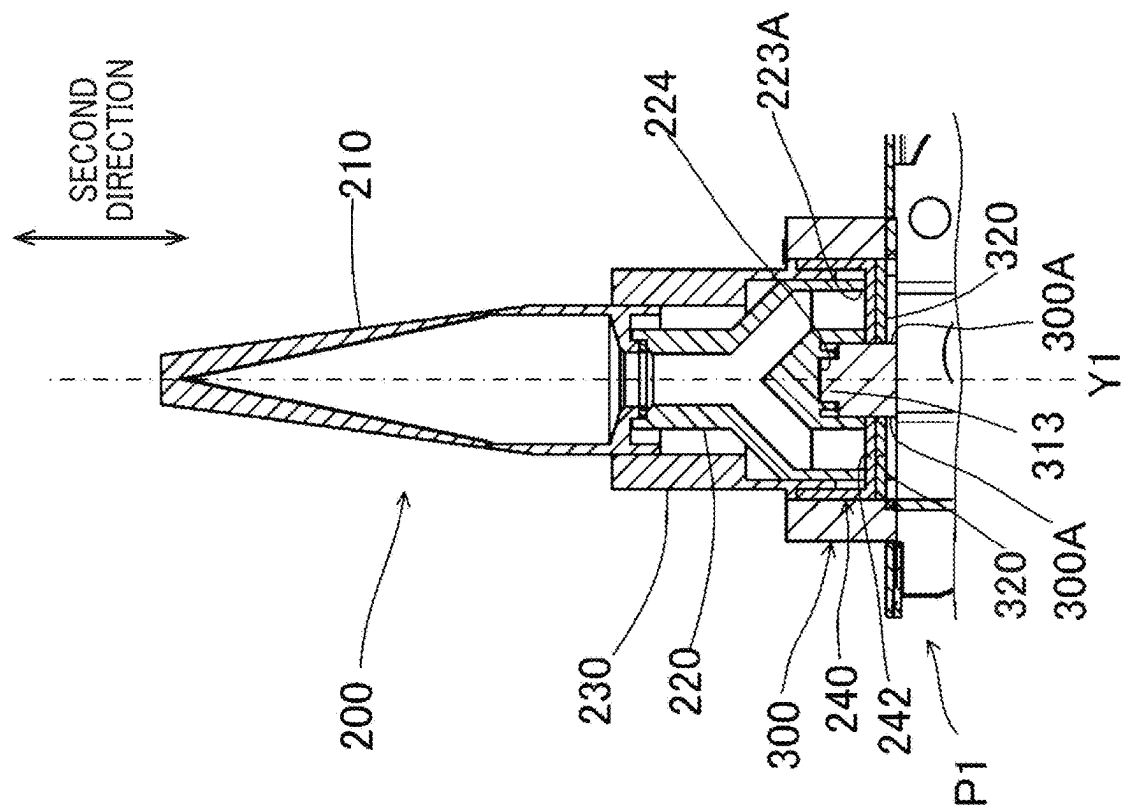
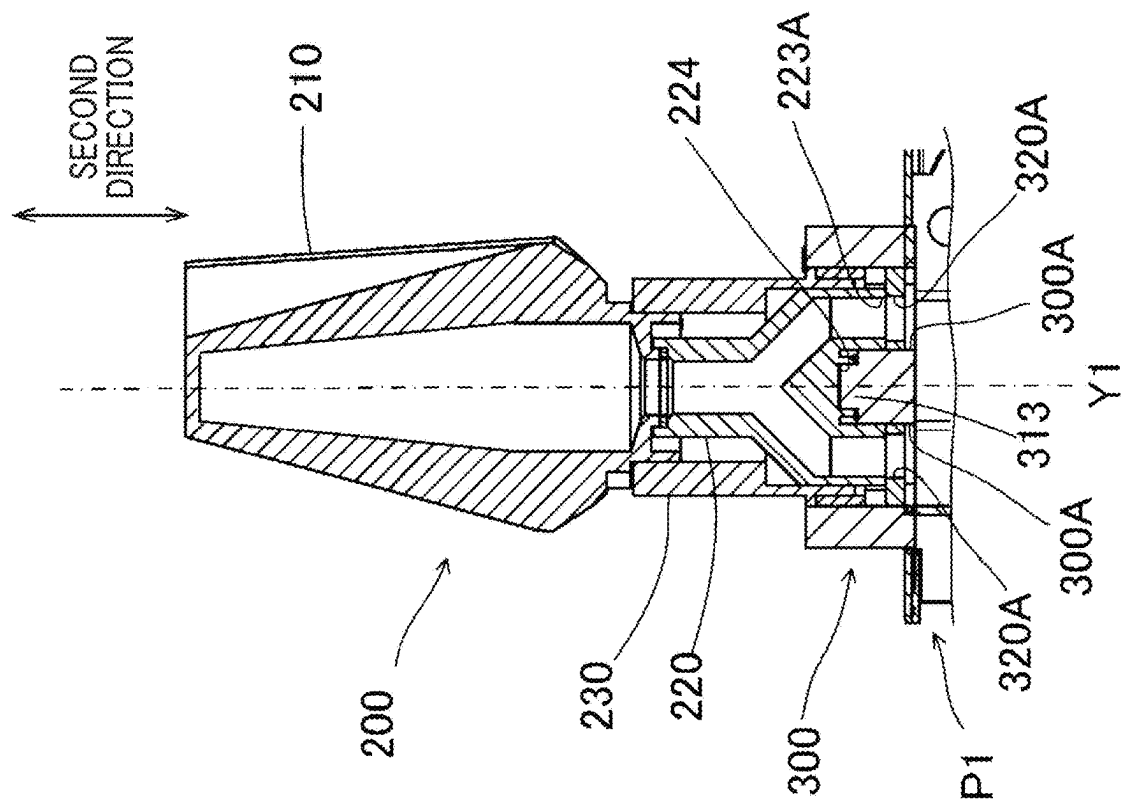
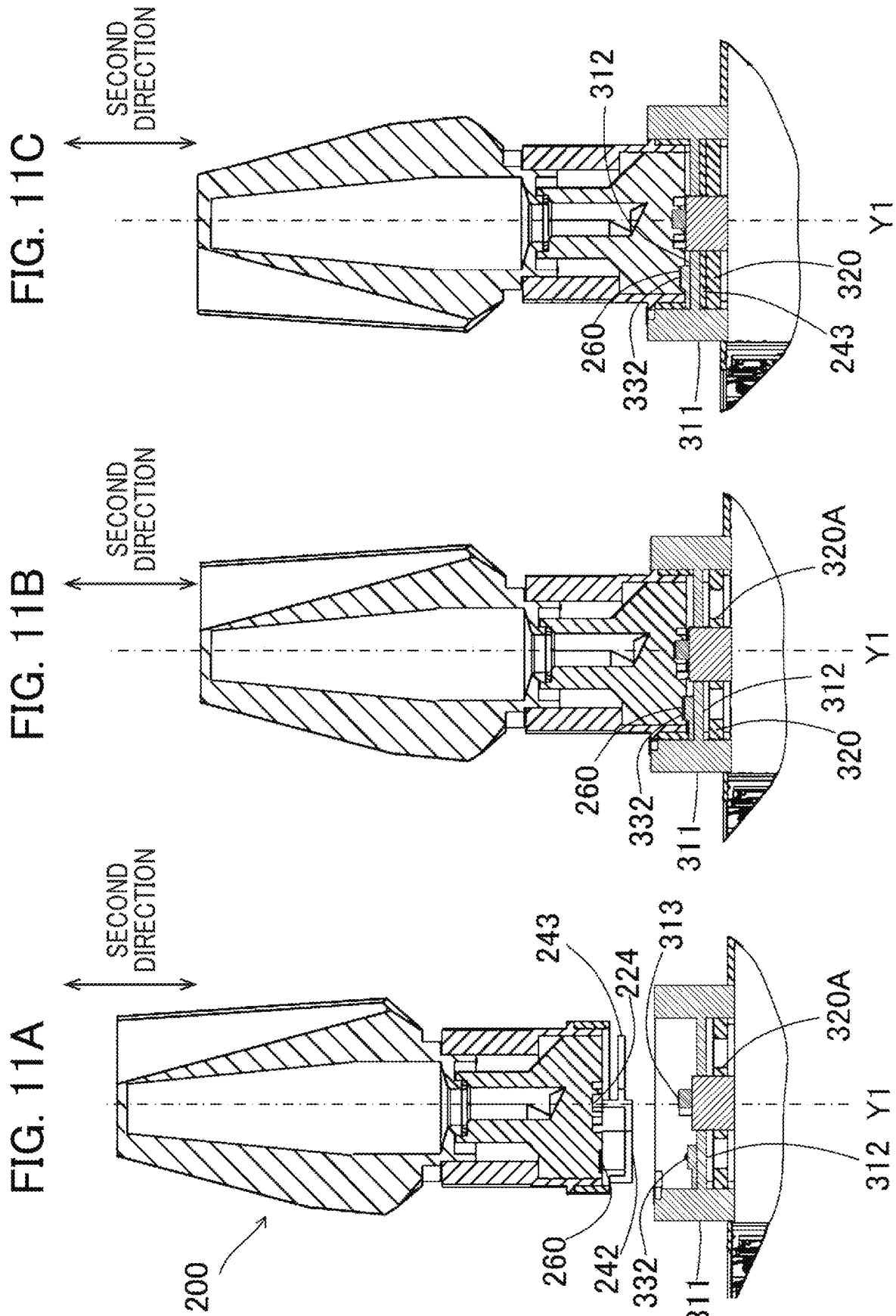


FIG. 10B





**IMAGE FORMING APPARATUS INCLUDING
PROCESS CARTRIDGE, AND TONER
CARTRIDGE HAVING TONER SHUTTER
PIVOTALLY MOVABLE IN A STATE WHERE
TONER CARTRIDGE IS ATTACHED TO
PROCESS CARTRIDGE**

REFERENCE TO RELATED APPLICATIONS

This is a by-pass continuation application of International Application No. PCT/JP2022/043899 filed on Nov. 29, 2022 which claims priority from Japanese Patent Application No. 2021-196723 filed on Dec. 3, 2021. The entire contents of the International Application and the priority application are incorporated herein by reference.

BACKGROUND ART

There has been conventionally known an image forming apparatus including a main housing to which a cartridge is attachable. The cartridge includes a developing roller and a developer container accommodating therein toner. In such a conventional image forming apparatus, replacing an old cartridge with a new cartridge is known as a system for replenishing the image forming apparatus with toner when toner in the old cartridge is used up. However, in such a system, replacement of the cartridge including the developing roller is performed even when the developing roller does not reach the end of a service life.

There has also been conventionally known another system in which a toner cartridge filled with toner for replenishment is attachable to a cartridge while the cartridge is attached to the main housing of the image forming apparatus for replenishing the image forming apparatus with toner. In this system, replenishment of the image forming apparatus with toner can be performed without replacing the developing roller.

SUMMARY

However, in the latter system, a toner memory storing therein information on the toner cartridge must be provided at the toner cartridge in order to enable the image forming apparatus to perform control operation in conformance with the attached toner cartridge.

According to one aspect, the present disclosure provides an image forming apparatus including: a main housing; a process cartridge; and a toner cartridge. The main housing has a first opening and a second opening. The main housing includes a main electrical contact. The process cartridge is attachable to and detachable from the main housing through the first opening. The process cartridge includes: a developing roller; a process housing; a first process electrical contact; and a second process electrical contact. The developing roller is rotatable about a developing roller axis extending in a first direction. The process housing has a toner inlet port and is configured to accommodate toner therein. The first process electrical contact is in contact with the main electrical contact in a state where the process cartridge is attached to the main housing. The second process electrical contact is electrically connected to the first process electrical contact. The toner cartridge is attachable to and detachable from the process cartridge through the second opening in a second direction different from the first direction in a state where the process cartridge is attached to the main housing. The toner cartridge includes: a toner housing; a toner memory; and a toner shutter. The toner

housing has a toner outlet port and is configured to accommodate toner therein. The toner memory includes a toner electrical contact. The toner memory stores therein information related to the toner cartridge. The toner shutter is configured to open and close the toner outlet port. In a state where the toner cartridge is attached to the process cartridge: a part of the toner housing is exposed to an outside of the main housing; the toner electrical contact is in contact with the second process electrical contact; and the toner shutter is pivotally movable about a toner shutter axis extending in the second direction relative to the second process electrical contact and the toner electrical contact between: a first position in which the toner shutter closes the toner outlet port; and a second position in which the toner shutter opens the toner outlet port.

With the above structure, in a state where the toner cartridge is attached to the process cartridge, the toner electrical contact can be electrically connected to the second process electrical contact of the process cartridge.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a cross-sectional view of a monochromatic printer as an example of the image forming apparatus.

FIG. 2A is a perspective view of the monochromatic printer in which a process cartridge is attached to a main housing, and illustrating a state where a replenishment cover is closed.

FIG. 2B is a perspective view of the monochromatic printer in which the process cartridge is attached to the main housing, and illustrating a state where the replenishment cover is open and a toner cartridge is not attached to the process cartridge.

FIG. 2C is a perspective view of the monochromatic printer in which the process cartridge is attached to the main housing, and illustrating a state where the replenishment cover is open and the toner cartridge is attached to the process cartridge.

FIG. 3A is a perspective view of the monochromatic printer in which a front cover and a top cover are open, and illustrating a state where the process cartridge is attached to the main housing.

FIG. 3B is a perspective view of the monochromatic printer in which the front cover and the top cover are open, and illustrating a state where the process cartridge is not attached to the main housing.

FIG. 3C is an enlarged perspective view of a main electrical contact.

FIG. 4A is a perspective view of the process cartridge as viewed from a side at which a first process electrical contact is positioned.

FIG. 4B is a perspective view of the process cartridge as viewed from a side at which a toner replenishing part is positioned.

FIG. 4C is a partial enlarged perspective view of the process cartridge illustrating a portion in the vicinity of a second process electrical contact positioned on the toner replenishing part.

FIG. 5A is a perspective view of the toner replenishing part, and illustrating a state where a process shutter is in a third position.

FIG. 5B is a perspective view of the toner replenishing part, and illustrating a state where the process shutter is in a fourth position.

FIG. 5C is a front view of the toner replenishing part, and illustrating a state where the process shutter is in the third position.

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FIG. 5D is a front view of the toner replenishing part, and illustrating a state where the process shutter is in the fourth position.

FIG. 5E is a cross-sectional view of the toner replenishing part taken along line VE-VE in FIG. 5C.

FIG. 6A is a perspective view of the toner cartridge, and illustrating a state where a toner shutter is in a first position.

FIG. 6B is a perspective view of the toner cartridge, and illustrating a state where the toner shutter is in a second position.

FIG. 7A is an exploded perspective view of the toner cartridge.

FIG. 7B is a cross-sectional view of the toner cartridge.

FIG. 8A is a view of the process cartridge and the toner cartridge, and illustrating a state prior to attaching the toner cartridge to the process cartridge.

FIG. 8B is a view of the process cartridge and the toner cartridge attached thereto, and illustrating a first attachment state.

FIG. 8C is a view of the process cartridge and the toner cartridge attached thereto, and illustrating a second attachment state in which the toner shutter has pivotally moved from the first attachment state.

FIG. 9A is a perspective view of the toner replenishing part.

FIG. 9B is a perspective view of the toner cartridge.

FIG. 10A is a cross-sectional view of the process cartridge and the toner cartridge attached thereto, and illustrating the first attachment state.

FIG. 10B is a cross-sectional view of the process cartridge and the toner cartridge attached thereto, and illustrating the second attachment state.

FIG. 11A is a cross-sectional view of the process cartridge and the toner cartridge, and illustrating a state prior to attaching the toner cartridge to the process cartridge.

FIG. 11B is a cross-sectional view of the process cartridge and the toner cartridge attached thereto, and illustrating the first attachment state.

FIG. 11C is a cross-sectional view of the process cartridge and the toner cartridge attached thereto, and illustrating the second attachment state.

DESCRIPTION

Hereinafter, one embodiment of the present disclosure will be described while referring to the accompanying drawings.

1. Overview of Monochromatic Printer 1

As illustrated in FIGS. 1 through 2C, a monochromatic printer 1 as an example of the image forming apparatus includes a main housing 10, a controller 70, a process cartridge P1, and a toner cartridge 200.

As illustrated in FIGS. 1 and 3A, the process cartridge P1 includes a process housing 31, a developing roller 32, a supply roller 33, an agitator 34, a cartridge handle 35, a photosensitive drum 41, and a charge roller 42.

In the following description, a direction in which a rotation axis (a developing axis X1) of a developing roller 22 of a developing cartridge 20 extends will be referred to as “first direction”. Further, a direction in which the toner cartridge 200 is attachable to and detachable from the process cartridge P1 in a state where the process cartridge P1 is attached to the main housing 10 of the monochromatic printer 1 will be referred to as “second direction”. Still

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further, a direction perpendicular to both the first direction and the second direction will be referred to as “third direction”.

The process housing 31 is configured to accommodate toner therein. The process housing 31 has toner inlet ports 300A.

The developing roller 32 is configured to supply toner accommodated in the process housing 31 to an outer circumferential surface of the photosensitive drum 41. The developing roller 32 is rotatable about the developing roller axis X1 extending in the first direction. The developing roller 32 is positioned at one end in the third direction of the process housing 31 in a state where the process cartridge P1 is attached to the main housing 10.

The supply roller 33 is rotatable about a supply roller axis extending in the first direction. The supply roller 33 is configured to supply toner to an outer circumferential surface of the developing roller 32.

The agitator 34 is rotatable about an agitator axis extending in the first direction. The agitator 34 is configured to agitate toner in the process housing 31.

The photosensitive drum 41 is configured to carry on the outer circumferential surface an electrostatic latent image formed by an exposure unit (not illustrated), and also to carry on the outer circumferential surface a toner image formed by toner supplied from the developing roller 32 to the carried electrostatic latent image. The photosensitive drum 41 is also configured to transfer the carried toner image onto a sheet S when the sheet S is nipped between the photosensitive drum 41 and a transfer roller 93 (described later).

The charge roller 42 is configured to charge the outer circumferential surface of the photosensitive drum 41 by contacting the photosensitive drum 41 while a voltage is applied thereto.

A fixing unit 80 includes a heating unit 81, and a pressure unit 82. The fixing unit 80 is configured to fix the toner image to the sheet S when the sheet S having the toner image transferred thereon is nipped between the heating unit 81 and the pressure unit 82.

A sheet conveying unit 90 includes a sheet accommodating unit 91, a pick-up roller 92, the transfer roller 93, and discharge rollers 94. The sheet accommodating unit 91 is configured to accommodate therein sheets S. The pick-up roller 92 is configured to convey the sheet S from the sheet accommodating part 91 toward the photosensitive drum 41. The transfer roller 93 is configured to convey the sheet S toward the fixing unit 80 while nipping the sheet S in cooperation with the photosensitive drum 41. The discharge rollers 94 are configured to discharge the sheet S passed through the fixing unit 80 out of the main housing 10. The sheet S discharged out of the main housing 10 is to be mounted on a discharge tray T.

As illustrated in FIGS. 2B and 2C, the toner cartridge 200 is attachable to the process cartridge P1 in a state where the process cartridge P1 is attached to the main housing 10. When attached to the process cartridge P1, the toner cartridge 200 can supply toner to the process cartridge P1.

The controller 70 includes, for example, a CPU, a RAM, a ROM, and an input/output circuit. The controller 70 is configured to control operations in the monochromatic printer 1 by executing arithmetic processing on the basis of a program stored in the ROM and the like. Further, the controller 70 is configured to control operations in the monochromatic printer 1 on the basis of information stored in a process memory (not illustrated) of the process cartridge P1 attached to the main housing 10, information stored in a

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toner memory 260 (FIG. 6A, described later) of the toner cartridge 200 attached to the process cartridge P1, and the like.

2. Details of Main Housing 10

As illustrated in FIGS. 2A through 3C, the main housing 10 includes a front cover 11, a top cover 12, a replenishment cover 13, and a main electrical contact 14. Further, the main housing 10 has a first opening 10A, and a second opening 10B.

As illustrated in FIGS. 3A and 3B, the front cover 11 is configured to open and close a part of the first opening 10A. A user can place sheets S in the sheet accommodating unit 91 in a state where the front cover 11 is open.

The top cover 12 is configured to open and close another part of the first opening 10A. In a state where the top cover 12 is closed, the top cover 12 constitutes the discharge tray T. The process cartridge P1 is attachable to and detachable from the main housing 10 through the first opening 10A. Hence, the process cartridge P1 is attachable to and detachable from the main housing 10 in a state where the top cover 12 is open.

As described above, the process cartridge P1 includes the cartridge handle 35. The cartridge handle 35 is positioned at another end in the third direction of the process housing 31. The user grips the cartridge handle 35 when attaching and detaching the process cartridge P1 to and from the main housing 10.

The replenishment cover 13 is positioned at one end in the second direction of the main housing 10. The replenishment cover 13 is configured to open and close the second opening 10B. As illustrated in FIG. 2A, the replenishment cover 13 constitutes the discharge tray T in a state where the replenishment cover 13 closes the second opening 10B. As illustrated in FIG. 2B, the replenishment cover 13 allows a toner replenishing part 300 (described later) of the process cartridge P1 to be exposed to an outside in a state where the replenishment cover 13 opens the second opening 10B.

As illustrated in FIGS. 2B and 2C, the toner cartridge 200 is attachable to and detachable from the process cartridge P1 in the second direction through the second opening 10B in a state where the process cartridge P1 is attached to the main housing 10. That is, the toner cartridge 200 is attachable to the process cartridge P1 in a state where the replenishment cover 13 is open.

As illustrated in FIGS. 3A through 3C, the main electrical contact 14 is positioned below the process cartridge P1 in a state where the process cartridge P1 is attached to the main housing 10. The main electrical contact 14 is in contact with a first process electrical terminal 331 (FIG. 4A, described later) in a state where the process cartridge P1 is attached to the main housing 10. The main electrical contact 14 is electrically connected to the controller 70.

3. Particular Structure of Process Cartridge P1 in Relation to Toner Cartridge 200

3-1. Overview of Process Cartridge P1

As illustrated in FIGS. 4A through 4C, the process cartridge P1 includes the toner replenishing part 300, and a process electrical relay part 330.

The toner replenishing part 300 is a part to which the toner cartridge 200 is attachable. As illustrated in FIG. 4B, the toner replenishing part 300 is positioned at one end in the second direction of the process housing 31. Toner in the

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toner cartridge 200 is supplied to the process housing 31 through the toner inlet ports 300A in a state where the toner cartridge 200 is attached to the toner replenishing part 300.

The process electrical relay part 330 electrically connects the toner memory 260 of the toner cartridge 200 and the main electrical contact 14 of the main housing 10 to each other in a state where the toner cartridge 200 is attached to the toner replenishing part 300.

3-2. Details of Process Electrical Relay Part 330

The process electrical relay part 330 includes the first process electrical terminal 331, a second process electrical terminal 332, and a harness 333.

As illustrated in FIG. 4A, the first process electrical terminal 331 is positioned opposite the toner replenishing part 300 in the second direction. In other words, the first process electrical terminal 331 is positioned at another end in the second direction of the process housing 31. The first process electrical terminal 331 is in contact with the main electrical contact 14 in a state where the process cartridge P1 is attached to the main housing 10.

The second process electrical terminal 332 is positioned at the toner replenishing part 300. In other words, the second process electrical terminal 332 is positioned at the one end in the second direction of the process housing 31. The second process electrical terminal 332 is in contact with a toner electrical contact 261 (FIG. 6A, described later) of the toner memory 260 in a state where the toner cartridge 200 is attached to the process cartridge P1.

The harness 333 electrically connects the first process electrical terminal 331 and the second process electrical terminal 332 to each other. Accordingly, the second process electrical terminal 332 is electrically connected to the first process electrical terminal 331. The harness 333 extends along an opposite end in the third direction of the process housing 31 from the developing roller 32 to electrically connect the first process electrical terminal 331 and the second process electrical terminal 332 to each other. In other words, at least a part of the harness 333 is positioned at the other end in the third direction of the process housing 31.

Note that the first process electrical terminal 331 and the second process electrical terminal 332 are positioned closer to the other end in the third direction of the process housing 31 than to the one end in the third direction of the process housing 31.

3-3. Details of Toner Replenishing Part 300

As illustrated in FIGS. 5A through 5E, the toner replenishing part 300 includes a main body part 310, and a process shutter 320. The process shutter 320 is pivotally movable relative to the main body part 310 to open and close the toner inlet ports 300A.

The main body part 310 includes a hollow cylindrical part 311, process retaining plates 312, and a protrusion 313.

The hollow cylindrical part 311 protrudes in the second direction from the process housing 31 and extends in the second direction. The hollow cylindrical part 311 covers at least a part of an outer peripheral surface of a toner shutter 240 (FIGS. 6A and 6B, described later) in a state where the toner cartridge 200 is attached to the process cartridge P1. The hollow cylindrical part 311 includes a first mark 311A and a second mark 311B, and has a guide groove 311C.

The first mark 311A is a protrusion protruding in the second direction. The first mark 311A has a triangular shape when viewed in the second direction.

The second mark **311B** is a protrusion protruding in the second direction. The second mark **311B** has a triangular shape when viewed in the second direction. The second mark **311B** is positioned downstream relative to the first mark **311A** in a pivot direction **R** of the process shutter **320** when the process shutter **320** pivotally moves from a process closed position to a process open position.

The guide groove **311C** is a groove recessed in the second direction. The guide groove **311C** is configured to guide a handle protrusion **233** (FIG. 7A, described later) of the toner cartridge **200**. The guide groove **311C** extends between the first mark **311A** and the second mark **311B** in a pivot direction of the process shutter **320**.

The process retaining plates **312** protrude radially inward from an inner peripheral surface of the hollow cylindrical part **311**.

Note that the second process electrical terminal **332** is positioned on one of the process retaining plates **312**. The second process electrical terminal **332** faces in the second direction.

The protrusion **313** is positioned at a diametrical center of the hollow cylindrical part **311**. The protrusion **313** is provided at a position overlapped with a process shutter axis **Y2** which is a pivot axis of the process shutter **320**. The protrusion **313** protrudes in the second direction. Specifically, the protrusion **313** protrudes in a direction away from the process shutter **320** in the second direction. The protrusion **313** has a substantially rectangular shape when viewed in the second direction. However, the protrusion **313** may have a shape other than a rectangular shape as long as the shape of the protrusion **313** is determined so as to be in conformance with a shape of a groove **224** (FIGS. 7A and 7B, described later) of the toner cartridge **200**.

The process shutter **320** is pivotally movable about the process shutter axis **Y2** extending in the second direction. The process shutter **320** has a disc like shape, and is positioned in an inner space defined by the hollow cylindrical part **311**. The process shutter **320** includes first stepped parts **321A**, and second stepped parts **321B**. The first stepped parts **321A** and the second stepped parts **321B** are configured to be pressed by the toner shutter **240** (described later) of the toner cartridge **200** attached to the process cartridge **P1** for pivotally moving the process shutter **320** about the process shutter axis **Y2**.

As described above, the process shutter **320** is configured to open and close the toner inlet ports **300A**. Specifically, the process shutter **320** is pivotally movable between a third position (a position illustrated in FIGS. 5A and 5C) in which the process shutter **320** closes the toner inlet ports **300A** and a fourth position (a position illustrated in FIGS. 5B and 5D) in which the process shutter **320** opens the toner inlet ports **300A**.

The process shutter **320** has shutter openings **320A**. In a state where the process shutter **320** is in the fourth position, the shutter openings **320A** are in communication with the respective toner inlet ports **300A**. Hence, the toner inlet ports **300A** are exposed to the outside through the respective shutter openings **320A**.

4. Details of Toner Cartridge **200**

4-1. Outline of Toner Cartridge **200**

As illustrated in FIGS. 6A through 7B, the toner cartridge **200** includes a toner housing **201**, a handle **230**, the toner shutter **240**, a seal member **250**, and the toner memory **260**.

The toner housing **201** is configured to accommodate toner therein. The toner housing **201** has toner outlet ports **223A**. More specifically, the toner housing **201** includes a toner container **210** accommodating toner therein, and a nozzle **220** having the toner outlet ports **223A**.

The toner container **210** has a toner container opening **210A** allowing toner accommodated in the toner container **210** to be discharged out of the toner container **210** there-through. The toner container **210** is made of resin such as ABS (acrylonitrile butadiene styrene), BS (butadiene styrene), and the like. However, the toner container **210** may be made of other material. For example, the toner container **210** may be a bag made from flexible plastic, or a paper container. A part of the toner container **210** is exposed to the outside of the main housing **10** in a state where the toner cartridge **200** is attached to the process cartridge **P1**. In other words, a part of the toner housing **201** is exposed to the outside of the main housing **10** in a state where the toner cartridge **200** is attached to the process cartridge **P1**.

As described above, the nozzle **220** has the toner outlet ports **223A**. The nozzle **220** is configured to allow toner accommodated in the toner container **210** to be discharged out of the toner cartridge **200** through the toner outlet ports **223A**.

The handle **230** has a hollow cylindrical shape. The handle **230** covers an outer peripheral surface of the nozzle **220**. In other words, the nozzle **220** is positioned inside the hollow cylindrical handle **230**. The handle **230** is positioned between the toner container **210** and the toner shutter **240** in the second direction. In other words, the toner container **210** is positioned at one end in the second direction of the handle **230**, and the toner shutter **240** is positioned at another end in the second direction of the handle **230**.

The toner shutter **240** is positioned on the opposite side of the nozzle **220** from the toner container **210** in the second direction. In a state where the toner cartridge **200** is attached to the process cartridge **P1**, the toner shutter **240** is pivotally movable about a toner shutter axis **Y1** extending in the second direction between a first position (a position illustrated in FIG. 6A) in which the toner shutter **240** closes the toner outlet ports **223A** and a second position (a position illustrated in FIG. 6B) in which the toner shutter **240** opens the toner outlet ports **223A**.

Note that the toner shutter axis **Y1** and the process shutter axis **Y2** coincide with each other in a state where the toner cartridge **200** is attached to the process cartridge **P1**. In other words, the toner shutter **240** and the process shutter **320** are coaxially pivotally movable in a state where the toner cartridge **200** is attached to the process cartridge **P1**.

As illustrated in FIGS. 7A and 7B, the seal member **250** is nipped between the toner container **210** and the nozzle **220** having a communication port **221A**. The seal member **250** restrains toner directed from the toner container opening **210A** toward the communication port **221A** from leaking out of the toner cartridge **200** through a gap between the toner container **210** and the nozzle **220**.

The toner memory **260** is a storage medium configured to store therein information related to the toner cartridge **200**. For example, the toner memory **260** stores therein information such as: an amount of toner accommodated in the toner cartridge **200**; a serial number of the toner cartridge **200**; the date of manufacture of the toner cartridge **200**; the color of toner accommodated in the toner cartridge **200**; and the country in which the toner cartridge **200** is sold; and the like. The toner memory **260** is positioned on the nozzle **220**. The toner memory **260** includes the toner electrical contact **261**. The toner electrical contact **261** is positioned on the nozzle

220. The toner electrical contact **261** faces in the second direction. Specifically, the toner electrical contact **261** faces in a direction away from the toner container **210** in the second direction.

4-2. Details of Nozzle **220**

The nozzle **220** includes a first cylindrical part **221**, a second cylindrical part **222**, and protruding parts **223**. Further, the nozzle **220** has the groove **224**.

The first cylindrical part **221** has the communication port **221A** which is in communication with the toner container opening **210A**.

The second cylindrical part **222** has a bottom surface **222A**, and an attachment groove **222B** recessed from the bottom surface **222A** in the second direction. The toner memory **260** is attached to the attachment groove **222B**.

The protruding parts **223** protrude from the bottom surface **222A** in a direction away from the first cylindrical part **221** in the first direction. The protruding parts **223** have the respective toner outlet ports **223A**, which are in communication with the communication port **221A**. Accordingly, the nozzle **220** allows toner in the toner container **210** to be discharged out through the toner container opening **210A**, the communication port **221A**, and the toner outlet ports **223A**.

In the present embodiment, the nozzle **220** includes two of the protruding parts **223**, and the toner outlet port **223A** are formed in the respective two protruding parts **223**.

The groove **224** is recessed in a direction toward the toner container **210** in the second direction. The groove **224** is formed at a position overlapped with the toner shutter axis **Y1**. That is, the groove **224** is aligned with the toner shutter axis **Y1**. The groove **224** has a substantially rectangular shape when viewed in the first direction. Note that the shape of the groove **224** need not be a rectangular shape, but may be a triangular shape or an elliptical shape when viewed in the first direction as long the protrusion **313** of the process shutter **320** can be inserted in the groove **224** to avoid pivotal movement of the protrusion **313** relative to the groove **224**.

4-3. Details of Handle **230**

The handle **230** has a hollow cylindrical shape and covers the outer peripheral surface of the nozzle **220**. The toner container **210** and the toner shutter **240** are attached to the handle **230**. With this configuration, as illustrated in FIGS. **6A** and **6B**, the toner container **210** and the toner shutter **240** pivotally move together with the handle **230** as the handle **230** pivotally moves about the toner shutter axis **Y1**. The handle **230** includes a grip part **231**, an engagement part **232**, and the handle protrusion **233**.

The handle protrusion **233** has an outer peripheral surface having a polygonal shape. The user grips the grip part **231** when pivotally moving the handle **230**. The toner container **210** is attached to an inner portion of the grip part **231**.

The engagement part **232** is positioned on the opposite side of the grip part **231** from the toner container **210** in the second direction. The engagement part **232** is a part to which the toner shutter **240** is attached.

The handle protrusion **233** is a protrusion that protrudes radially outward from the engagement part **232**. The handle protrusion **233** is a visible part for the user to visually recognize a pivot angle of the handle **230**.

4-4. Details of Toner Shutter **240**

The toner shutter **240** includes a hollow cylindrical part **241**, shielding plates **242**, and toner retaining plates **243**.

The hollow cylindrical part **241** covers an outer peripheral surface of the engagement part **232** of the handle **230**. The hollow cylindrical part **241** has one end in the second direction in contact with the handle **230**.

The shielding plates **242** are positioned at another end in the second direction of the hollow cylindrical part **241**. Each of the shielding plates **242** extends from the hollow cylindrical part **241** toward a diametrical center of the hollow cylindrical part **241**. Each of the shielding plates **242** closes a corresponding one of the toner outlet ports **223A** in a state where the toner shutter **240** is in the first position.

Each of the shielding plates **242** has a first end **242A**, and a second end **242B**. With respect to a pivot direction **Q** (see FIG. **6B**) of the toner shutter **240** when the toner shutter **240** pivotally moves from the first position toward the second position, the first end **242A** serves as a downstream end of the shielding plate **242**, and the second end **242B** serves as an upstream end of the shielding plate **242**.

The toner retaining plates **243** are positioned at the other end in the second direction of the hollow cylindrical part **241**. The toner retaining plates **243** are positioned closer to the toner container **210** than the shielding plates **242** are to the toner container **210** in the second direction. The toner shutter **240** defines gaps between the toner retaining plates **243** and the hollow cylindrical part **241** to allow the process retaining plates **312** to be inserted therein.

5. Attachment Operation of Toner Cartridge **200** to Toner Replenishing Part **300**

5-1. Outline of Attachment Operation

Next, an attachment operation for attaching the toner cartridge **200** to the toner replenishing part **300** will be described with reference to FIGS. **8A** through **11C**.

In a state prior to attaching the toner cartridge **200** to the toner replenishing part **300** illustrated in FIG. **8A**, the toner shutter **240** is in the first position (see FIG. **6A**). Also, in a state prior to attaching the toner cartridge **200** to the toner replenishing part **300**, the process shutter **320** is in the third position (see FIG. **5A**).

From a state illustrated in FIG. **8A**, the user inserts the toner shutter **240** of the toner cartridge **200** into the hollow cylindrical part **311** of the toner replenishing part **300** as illustrated in FIG. **8B**. At this time, the user aligns a distal end of the handle protrusion **233** with the first mark **311A** when inserting the toner shutter **240**. In the following description, a state of the toner cartridge **200** and the toner replenishing part **300** illustrated in FIG. **8B** will be referred to as "first attachment state". FIGS. **10A** and **11B** also illustrate the first attachment state.

Then, from the state illustrated in FIG. **8B**, the user pivotally moves the toner cartridge **200** relative to the process cartridge **P1** as illustrated in FIG. **8C**. Specifically, the user grips the grip part **231** and pivotally moves the handle **230** relative to the process housing **31** until the distal end of the handle protrusion **233** moves from the first mark **311A** to reach the second mark **311B**. In the following description, a state of the toner cartridge **200** and the toner replenishing part **300** illustrated in FIG. **8C** will be referred to as "second attachment state". Note that FIGS. **10B** and **11C** also illustrate the second attachment state.

Next, a relationship between components of the toner cartridge **200** and components of the toner replenishing part **300** will be described.

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5-2. Relationship Between Toner Shutter 240 and Process Shutter 320

As illustrated in FIG. 9B, the toner shutter 240 has a first surface S1, and a second surface T1. Both the first surface S1 and the second surface T1 face in the second direction. In the meantime, the nozzle 220 has a third surface U1. The third surface U1 is exposed to the outside in the second direction in a state where the toner shutter 240 is in the first position.

Further, as illustrated in FIG. 9A, the process shutter 320 has a fourth surface S2 and a fifth surface T2 those facing in the second direction. Further, the main body part 310 has a sixth surface U2 facing in the second direction.

In the first attachment state, the toner shutter 240 and the process shutter 320 provide such a relationship that the first surface S1 faces the fourth surface S2 in the second direction, the second surface T1 faces the fifth surface T2 in the second direction, and the third surface U1 faces the sixth surface U2 in the second direction.

Further, in the first attachment state, the first ends 242A face the corresponding first stepped parts 321A in the pivot direction of the toner shutter 240, and the second ends 242B face the corresponding second stepped parts 321B in the pivot direction of the toner shutter 240.

As the toner cartridge 200 pivotally moves such that the toner cartridge 200 and the toner replenishing part 300 shift from the first attachment state to the second attachment state, the first ends 242A of the shielding plates 242 press the corresponding first stepped parts 321A to pivotally move the process shutter 320. In other words, the toner shutter 240 presses the process shutter 320 such that the process shutter 320 pivotally moves from the third position to the fourth position when the toner shutter 240 pivotally moves from the first position to the second position while the toner cartridge 200 is attached to the process cartridge P1.

Also, as the toner cartridge 200 pivotally moves such that the toner cartridge 200 and the toner replenishing part 300 shift from the second attachment state to the first attachment state, the second ends 242B of the shielding plates 242 press the corresponding second stepped parts 321B to pivotally move the process shutter 320. In other words, the toner shutter 240 presses the process shutter 320 such that the process shutter 320 pivotally moves from the fourth position to the third position when the toner shutter 240 pivotally moves from the second position to the first position while the toner cartridge 200 is attached to the process cartridge P1.

In this way, the process shutter 320 is pivotally movable together with the toner shutter 240 in a state where the toner cartridge 200 is attached to the process cartridge P1.

5-3. Relationship Between Groove 224 and Protrusion 313

As illustrated in FIG. 10A, in the first attachment state, the protrusion 313 is inserted in the groove 224. In other words, in a state where the toner cartridge 200 is attached to the process cartridge P1, the protrusion 313 is inserted in the groove 224. The nozzle 220 is restrained from pivotally moving relative to the toner replenishing part 300 by the insertion of the protrusion 313 in the groove 224. This configuration enables the pivotal movement of the toner shutter 240 relative to the nozzle 220 between the first position and the second position while the protrusion 313 is inserted in the groove 224, as illustrated in FIGS. 10A and 10B.

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Note that the toner shutter 240 pivotally moves together with the handle 230 relative to the process housing 31 in accordance with the pivotal movement of the handle 230. On the other hand, the nozzle 220 does not pivotally move relative to the process housing 31. Therefore, the handle 230 is pivotally movable together with the toner shutter 240 about the toner shutter axis Y1 relative to the nozzle 220.

Further, the toner container 210 pivotally moves together with the handle 230 relative to the process housing 31 in accordance with the pivotal movement of the handle 230. Therefore, the toner container 210 is pivotally movable about the toner shutter axis Y1 relative to the nozzle 220 in a state where the toner cartridge 200 is attached to the process cartridge P1.

5-4. Relationship Between Toner Outlet Ports 223A and Toner Inlet Ports 300A

In the first attachment state illustrated in FIG. 10A, the toner shutter 240 closes the toner outlet ports 223A, and the process shutter 320 closes the toner inlet ports 300A. In other words, in the first attachment state, the toner shutter 240 is in the first position and the process shutter 320 is in the third position. Hence, toner is not supplied to the process housing 31 in the first attachment state.

Then, as the toner cartridge 200 pivotally moves, the toner cartridge 200 and the toner replenishing part 300 shift from the first attachment state illustrated in FIG. 10A to the second attachment state illustrated in FIG. 10B. At this time, the toner shutter 240 and the process shutter 320 pivotally moves together.

In the second attachment state illustrated in FIG. 10B, the toner shutter 240 opens the toner outlet ports 223A, and the process shutter 320 opens the toner inlet ports 300A. In other words, the toner shutter 240 is in the second position and the process shutter 320 is in the fourth position in the second attachment state. As a result, the toner outlet ports 223A are in communication with the corresponding toner inlet ports 300A through the shutter openings 320A. In the second attachment state, the user can replenish the process cartridge P1 with toner accommodated in the toner cartridge 200.

5-5. Relationship Between Toner Electrical Contact 261 and Second Process Electrical Terminal 332

In the first attachment state illustrated in FIG. 11B, the toner electrical contact 261 is in contact with the second process electrical terminal 332. Therefore, the toner electrical contact 261 is electrically connected to the second process electrical terminal 332.

The toner electrical contact 261 does not pivotally move since the toner electrical contact 261 is positioned on the nozzle 220. Further, the second process electrical terminal 332 does not pivotally move since the second process electrical terminal 332 is positioned on one of the process retaining plates 312 of the main body part 310. Hence, as illustrated in FIGS. 11B and 11C, the toner shutter 240 is pivotally movable between the first position and the second position relative to the second process electrical terminal 332 and the toner electrical contact 261.

The toner electrical contact 261 is electrically connected to the main electrical contact 14 through the second process electrical terminal 332, the harness 333, and the first process electrical terminal 331. Hence, the controller 70 of the monochromatic printer 1 can read information stored in the toner memory 260 through the main electrical contact 14.

5-6. Relationship Between Toner Retaining Plates 243 and Process Retaining Plates 312

As the toner cartridge 200 and the toner replenishing part 300 shift from the first attachment state to the second attachment state due to the pivotal movement of the toner cartridge 200, the toner shutter 240 pivotally moves from the first position to the second position to move the toner retaining plates 243 to positions immediately below the corresponding process retaining plates 312, as illustrated in FIG. 11C. As a result, in a state where the toner shutter 240 is in the second position, the toner retaining plates 243 are positioned between the corresponding process retaining plates 312 and the process shutter 320 in the second direction. Further, in a state where the toner shutter 240 is in the second position, the process retaining plates 312 are positioned between the corresponding toner retaining plates 243 and the nozzle 220 in the second direction.

In other words, in a state where the toner cartridge 200 is attached to the process cartridge P1 and the toner shutter 240 is in the second position, the process retaining plates 312 are positioned between the corresponding toner retaining plates 243 and the toner housing 201 in the second direction. With this configuration, the process retaining plates 312 restrain the toner retaining plates 243 from moving in the second direction to thereby restraining unintentional detachment of the toner cartridge 200 from the toner replenishing part 300.

Note that, in the second attachment state, the second process electrical terminal 332 is positioned on the opposite side of the process retaining plates 312 from the toner retaining plates 243 in the second direction. Hence, the toner retaining plates 243 do not interfere with the second process electrical terminal 332 when the toner retaining plates 243 move to the positions below the corresponding process retaining plates 312 in accordance with the pivotal movement of the toner shutter 240. Accordingly, the detachment of the toner cartridge 200 can be retained while the second process electrical terminal 332 and the toner electrical contact 261 are maintained contact with each other.

5-7. Summary

As described above, when the toner cartridge 200 and the toner replenishing part 300 are in the first attachment state, the toner shutter 240 is in the first position and the process shutter 320 is in the third position. Hence, replenishment of the process cartridge P1 with toner is not performed in the first attachment state. On the other hand, when the toner cartridge 200 and the toner replenishing part 300 are in the second attachment state, the toner shutter 240 is in the second position and the process shutter 320 is in the fourth position. Accordingly, toner can be supplied from the toner cartridge 200 to the process cartridge P1 in the second attachment state.

The toner shutter 240 and the process shutter 320 pivotally move together as the toner cartridge 200 and the toner replenishing part 300 change between the first attachment state and the second attachment state. On the other hand, the toner electrical contact 261 and the second process electrical terminal 332 do not pivotally move even when the toner cartridge 200 and the toner replenishing part 300 change between the first attachment state and the second attachment state.

Hence, in a state where the toner cartridge 200 is attached to the process cartridge P1, the toner shutter 240 is pivotally movable relative to the second process electrical terminal 332 and the toner electrical contact 261 between the first

position in which the toner shutter 240 closes the toner outlet ports 223A and the second position in which the toner shutter 240 opens the toner outlet ports 223A. Accordingly, the toner electrical contact 261 and the second process electrical terminal 332 are maintained contact with each other in spite of the pivotal movement of the toner shutter 240 between the first position and the second position.

6. Advantageous Effects

According to the above-described embodiment of the present disclosure, in a state where the toner cartridge 200 is attached to the process cartridge P1, the toner shutter 240 is pivotally movable relative to the second process electrical terminal 332 and the toner electrical contact 261 between the first position in which the toner shutter 240 closes the toner outlet ports 223A and the second position in which the toner shutter 240 opens the toner outlet ports 223A. Hence, in a configuration in which the toner cartridge 200 includes the toner memory 260, the toner outlet ports 223A can be opened and closed while maintaining electrical contact between the second process electrical terminal 332 and the toner electrical contact 261.

Further, since the toner electrical contact 261 faces in the second direction, which is also a direction in which the toner cartridge 200 is attached to the process cartridge P1, wear of the toner electrical contact 261 when the toner electrical contact 261 contacts the second process electrical terminal 332 can be reduced in comparison with a configuration in which the toner electrical contact 261 faces in a direction different from the direction of the attachment of the toner cartridge 200 to the process cartridge P1.

Further, positioning of the toner cartridge 200 relative to the process cartridge P1 is fixed by the insertion of the protrusion 313 in the groove 224. Still further, since both the protrusion 313 and the groove 224 have substantially rectangular shape when viewed in the second direction, the protrusion 313 can restrain the nozzle 220 from pivotal moving relative to the process housing 31, thereby enabling only the toner shutter 240 to pivotally move to open and close the toner outlet ports 223A.

Further, the protrusion 313 does not interfere with the pivotal movement of the toner shutter 240 in a state where the protrusion 313 is inserted in the groove 224, since the groove 224 is at a position overlapped with the toner shutter axis Y1. Assuming that the groove 224 is positioned offset from the toner shutter axis Y1, attention is needed for a design of the toner outlet ports 223A, the toner shutter 240, and the groove 224 such that these parts are not positioned within a locus of the pivotal movement of the toner shutter 240. In contrast, according to the configuration in the above-described embodiment, relatively high degree of freedom in design can be realized since the groove 224 does not interfere with the pivotal movement of the toner shutter 240.

Further, the toner container 210 is pivotally movable relative to the nozzle 220. This configuration can restrain the user from erroneously pivotally moving the toner shutter 240 to the position in which the toner outlet ports 223A are opened. Specifically, assuming that the toner container 210 is not pivotally movable relative to the nozzle 220, the toner shutter 240 may pivotally move relative to the nozzle 220 to open the toner outlet ports 223A in a case where the user pivotally moves the toner shutter 240 while gripping the toner container 210 in a state where the toner cartridge 200 is not attached to the process cartridge P1.

In other words, according to the above configuration, the toner outlet ports 223A may be opened and closed just by the

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pivotal movement of the toner shutter **240** due to the configuration in which the toner container **210** and the nozzle **220** are not pivotally movable relative to each other. In contrast, according to the configuration in the above-described embodiment, the toner shutter **240** does not open and close the toner outlet ports **223A** even if the toner shutter **240** tries to pivotally move, since the toner container **210** is also pivotally movable relative to the nozzle **220**.

Further, since the handle **230** covers the outer peripheral surface of the nozzle **220**, fixing of the nozzle **220** can be restrained in a state other than the state where the toner cartridge **200** is attached to the process cartridge **P1**. Hence, opening and closing of the toner outlet ports **223A** by the toner shutter **240** can be restrained in a state other than the state where the toner cartridge **200** is attached to the process cartridge **P1**.

Further, the toner electrical contact **261** is positioned on the nozzle **220**. With this arrangement, stabilized contact between the second process electrical terminal **332** and the toner electrical contact **261** can be obtained since the nozzle **220** does not pivotally move by the pivotal movement of the toner shutter **240** in a state where the toner cartridge **200** is attached to the process cartridge **P1**.

Further, the process shutter **320**, which opens and closes the toner inlet ports **300A**, is pivotally movable relative to the toner electrical contact **261** and the second process electrical terminal **332** in a state where the toner cartridge **200** is attached to the process cartridge **P1**. Accordingly, the toner inlet ports **300A** can be opened and closed while the toner electrical contact **261** and the second process electrical terminal **332** are in contact with each other, thereby realizing stabilized contact between the toner electrical contact **261** and the second process electrical terminal **332**.

Further, the toner outlet ports **223A** and the toner inlet ports **300A** are opened concurrently since the toner shutter **240** and the process shutter **320** pivotally move together, whereby leakage of toner to the outside can be suppressed in a state other than a state where the replenishment of the process cartridge **P1** with toner is performed.

Further, the process shutter **320** pivotally moves due to a pressing force applied by the toner shutter **240**. Hence, there is no need to provide a separate component for pivotally moving the process shutter **320**.

The second process electrical terminal **332** does not interfere with the toner retaining plates **243** at the time of insertion of the toner retaining plates **243** into the gap between the process retaining plates **312** and the process shutter **320**. Hence, unintentional detachment of the toner cartridge **200** from the toner replenishing part **300** can be restrained while maintaining contact between the second process electrical terminal **332** and the toner electrical contact **261**.

Further, contact of the first process electrical terminal **331** with the main electrical contact **14** can be easily realized, since the first process electrical terminal **331** is positioned at the other end in the second direction of the process housing **31**. More specifically, the first process electrical terminal **331** can contact the main electrical contact **14** as a result of attachment of the process housing **31** to the main housing **10** through the first opening **10A** before closing the front cover **11** and the top cover **12**.

Further, since both the first process electrical terminal **331** and the second process electrical terminal **332** are positioned close to the other end in the third direction of the process housing **31**, a length of the harness **333** can be shortened. With this configuration, mixing of noise on signals in

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communication lines from the toner memory **260** to the controller **70** through the harness **333** of the process cartridge **P1** can be reduced.

Further, since the replenishment cover **13** is provided to open and close the second opening **10B**, the replenishment cover **13** needs to be opened only when replenishing the process cartridge **P1** with toner from the toner cartridge **200**. Accordingly, when the toner cartridge **200** is not attached to the toner replenishing part **300**, foreign materials such as dust can be restrained from entering the main housing **10** and the toner inlet ports **300A** by closing the second opening **10B** with the replenishment cover **13**.

Further, the replenishment cover **13** constitutes the discharge tray **T** in a state where the replenishment cover **13** closes the second opening **10B**. Therefore, the process housing **31** and the main housing **10** can be made compact in comparison with a configuration in which the second opening **10B** and the replenishment cover **13** are positioned outside the discharge tray **T**.

7. Modifications

While the invention has been described in conjunction with various example structures outlined above and illustrated in the figures, various alternatives, modifications, variations, improvements, and/or substantial equivalents, whether known or that may be presently unforeseen, may become apparent to those having at least ordinary skill in the art. Accordingly, the example embodiments of the disclosure, as set forth above, are intended to be illustrative of the invention, and not limiting the invention. Various changes may be made without departing from the spirit and scope of the disclosure. Therefore, the disclosure is intended to embrace all known or later developed alternatives, modifications, variations, improvements, and/or substantial equivalents. Some specific examples of potential alternatives, modifications, or variations in the described invention are provided below:

For example, in the above-described embodiment, the toner container **210** is pivotally movable together with the toner shutter **240** in a state where the toner cartridge **200** is attached to the process cartridge **P1**. However, the toner shutter **240** need not be pivotally movable along with the pivotal movement of the toner container **210**.

Specifically, although the toner container **210** is fixed to the handle **230** in the above-described embodiment, the present disclosure need not be limited to this configuration. For example, the toner container **210** may be fixed to the nozzle **220**. In the latter case, the toner container **210** does not pivotally move, since the nozzle **220** does not pivotally move in a state where the toner cartridge **200** is attached to the process cartridge **P1**. With this configuration, since the toner container **210** does not pivotally move relative to the nozzle **220**, leakage of toner from a gap between the nozzle **220** and the toner container **210** is unlikely to occur.

Further, according to the above-described embodiment, the handle **230** and the toner shutter **240** are two components different from each other. However, the handle **230** and the toner shutter **240** may be formed as a single member.

Further, according to the above-described embodiment, the nozzle **220** of the toner cartridge **200** has the groove **224**, and the toner replenishing part **300** includes the protrusion **313**. However, configurations other than that described above may be employed. As an example, the toner replenishing part **300** may have a groove, and the nozzle **220** may include a protrusion that can be inserted in the groove of the toner replenishing part **300**.

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Further, although the groove **224** is positioned to be overlapped with the toner shutter axis **Y1** in the above-described embodiment, the present disclosure need not be limited to this configuration. For example, one of the two toner outlet ports **223A** may be omitted, and the groove **224** may be formed at a position in which the omitted outlet port **223A** were provided. In the latter case, the toner shutter **240** may be configured to open and close the single toner outlet port **223A** while avoiding overlapping relation between the locus of the toner shutter **240** and the groove **224**. Further, in this case, the protrusion **313** is positioned offset from the process shutter axis **Y2** so that the protrusion **313** can be inserted in the groove **224** in a state where the toner cartridge **200** is attached to the process cartridge **P1**.

Further, in the above-described embodiment, the first process electrical terminal **331** is positioned at the other end in the second direction of the process housing **31**. As a modification, the first process electrical terminal **331** may be positioned at the one end in the second direction of the process housing **31**. Alternatively, the first process electrical terminal **331** may be positioned at one end or another end in the first direction of the process housing **31**. Further, the harness **333** may be positioned at one end or another end in the first direction of the process housing **31**. Even in such a configuration, contact of the first process electrical terminal **331** with the main electrical contact **14** can be easily achieved, since the position of the first process electrical terminal **331** can be designed irrespective of the position of the second process electrical terminal **332**.

Further, according to the above-described embodiment, the process cartridge **P1** is attachable to and detachable from the main housing **10**. However, as a modification, the process cartridge **P1** may be modified to include a developing cartridge including a developing roller and a drum cartridge including a photosensitive drum, and the developing cartridge may be attachable to or detachable from the drum cartridge. Alternatively, a drum cartridge and a developing cartridge may be attachable to and detachable from the main housing **10** independently of each other.

Further, in the above-described embodiment, the monochromatic printer **1** serves as one example of the image forming apparatus. However, a color printer including process cartridges respectively corresponding to color of black, magenta, yellow, and cyan may also be available as the image forming apparatus. In the latter case, a toner cartridge corresponding to each process cartridge may be attachable thereto, or a toner cartridge corresponding to color of black whose toner is heavily used may only be attachable to the corresponding process cartridge.

Further, the parts and the components used in the embodiment and the modifications described above may be combined as appropriate to be implemented.

What is claimed is:

1. An image forming apparatus comprising:
 - a main housing having a first opening and a second opening, the main housing including:
 - a main electrical contact;
 - a process cartridge attachable to and detachable from the main housing through the first opening, the process cartridge including:
 - a developing roller rotatable about a developing roller axis extending in a first direction;
 - a process housing having a toner inlet port and configured to accommodate toner therein;
 - a first process electrical contact in contact with the main electrical contact in a state where the process cartridge is attached to the main housing; and

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- a second process electrical contact electrically connected to the first process electrical contact; and
- a toner cartridge attachable to and detachable from the process cartridge through the second opening in a second direction different from the first direction in a state where the process cartridge is attached to the main housing, the toner cartridge including:
 - a toner housing having a toner outlet port and configured to accommodate toner therein;
 - a toner memory including a toner electrical contact, the toner memory storing therein information related to the toner cartridge; and
 - a toner shutter configured to open and close the toner outlet port,
 wherein, in a state where the toner cartridge is attached to the process cartridge:
 - a part of the toner housing is exposed to an outside of the main housing;
 - the toner electrical contact is in contact with the second process electrical contact; and
 - the toner shutter is pivotally movable about a toner shutter axis extending in the second direction relative to the second process electrical contact and the toner electrical contact between:
 - a first position in which the toner shutter closes the toner outlet port; and
 - a second position in which the toner shutter opens the toner outlet port.

2. The image forming apparatus according to claim 1, wherein the toner electrical contact faces in the second direction.

3. The image forming apparatus according to claim 1, wherein the toner housing includes:

- a nozzle having the toner outlet port; and
- a toner container accommodating toner therein,

wherein the process housing includes:

- a protrusion protruding in the second direction,

wherein the nozzle has a groove in which the protrusion is inserted in a state where the toner cartridge is attached to the process cartridge, and

wherein the toner shutter is pivotally movable between the first position and the second position relative to the nozzle while the protrusion is inserted in the groove.

4. The image forming apparatus according to claim 3, wherein the groove is aligned with the toner shutter axis.

5. The image forming apparatus according to claim 4, wherein, in a state where the toner cartridge is attached to the process cartridge, the toner container is pivotally movable about the toner shutter axis together with the toner shutter relative to the nozzle.

6. The image forming apparatus according to claim 5, wherein the toner cartridge further includes:

- a handle covering an outer peripheral surface of the nozzle,

wherein the toner shutter is positioned at one end in the second direction of the handle,

wherein the toner container is positioned at another end in the second direction of the handle, and

wherein, in a state where the toner cartridge is attached to the process cartridge, the handle is pivotally movable about the toner shutter axis together with the toner shutter relative to the nozzle.

7. The image forming apparatus according to claim 3, wherein the toner electrical contact is positioned on the nozzle.

8. The image forming apparatus according to claim 7, wherein the toner shutter includes:

- a toner retaining plate,

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wherein the process housing further includes:

a process retaining plate, and

wherein, in a state where the toner cartridge is attached to the process cartridge and the toner shutter is in the second position:

the process retaining plate is positioned between the toner retaining plate and the toner housing in the second direction; and

the second process electrical contact is positioned on an opposite side of the process retaining plate from the toner retaining plate in the second direction.

9. The image forming apparatus according to claim 1, wherein the process housing includes:

a process shutter configured to open and close the toner inlet port, and

wherein, in a state where the toner cartridge is attached to the process cartridge, the process shutter is pivotally movable about a process shutter axis extending in the second direction relative to the second process electrical contact and the toner electrical contact between:

a third position in which the process shutter closes the toner inlet port; and

a fourth position in which the process shutter opens the toner inlet port.

10. The image forming apparatus according to claim 9, wherein, in a state where the toner cartridge is attached to the process cartridge, the process shutter is pivotally movable together with the toner shutter,

wherein, in a state where the toner shutter is in the first position, the process shutter is in the third position, and

wherein, in a state where the toner shutter is in the second position, the process shutter is in the fourth position.

11. The image forming apparatus according to claim 10, wherein, when the toner shutter pivotally moves from the first position to the second position in a state where the toner cartridge is attached to the process cartridge, the

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toner shutter presses the process shutter such that the process shutter pivotally moves from the third position to the fourth position.

12. The image forming apparatus according to claim 1, wherein the second process electrical contact is positioned at one end in the second direction of the process housing, and

wherein the first process electrical contact is positioned at another end in the second direction of the process housing.

13. The image forming apparatus according to claim 12, wherein the process cartridge further includes:

a harness electrically connecting the first process electrical contact and the second process electrical contact to each other,

wherein the developing roller is positioned at one end in a third direction of the process housing,

wherein at least a part of the harness is positioned at another end in the third direction of the process housing, and

wherein both the first process electrical contact and the second process electrical contact are positioned closer to the another end in the third direction of the process housing than to the one end in the third direction of the process housing.

14. The image forming apparatus according to claim 1, wherein the main housing further includes:

a replenishment cover positioned at one end in the second direction of the main housing and configured to open and close the second opening.

15. The image forming apparatus according to claim 14, wherein the main housing further includes:

a discharge tray on which a sheet discharged out of the main housing is to be mounted, and

wherein, in a state where the replenishment cover closes the second opening, the replenishment cover constitutes the discharge tray.

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